PROBLEM 11 | 1D Energy Landscapes

Conservative Force and Potential II

Original statement · Problems.pdf, p. 9

The potential energy function of a particle of mass m is:

$$V(x) = -\frac{c}{2}(x^2 - a^2)^2$$

where c and a are positive constants. Sketch this function, and describe the possible types of motion in the three cases:

(a) $E > 0$

(b) $E < -\frac{ca^4}{2}$

(c) $-\frac{ca^4}{2} < E < 0$

Full Analytical Solution (4-Phase Methodology)

PHASE 1

Physical Statement, Hypotheses & Parameters

- **Physical System:** A point particle of mass m moves along the Ox axis in a quartic potential landscape $V(x) = -\frac{c}{2}(x^2 - a^2)^2$.
- **Degree of Freedom:** 1 degree of freedom, $x(t) \in \mathbb{R}$.
- **Conservative Law:** No friction or external work ($W_{nc} = 0$), so $E = T + V(x) =$ constant.
- **Physical Constraint:** Real physical motion requires real velocity, hence $T = \frac{1}{2}mv^2 \geq 0$. The classically allowed regions are restricted to:

$$\{x \in \mathbb{R} : V(x) \leq E\}$$

- **Parameters:**
- $c > 0$: Potential scale constant $[\mathrm{J/m^4} = \mathrm{N/m^3}]$.
- $a > 0$: Characteristic length $[\mathrm{m}]$.

PHASE 2

Reference Frames, Vector Bases & Coordinate Geometry

- **Potential Expression:**

$$V(x) = -\frac{c}{2}(x^2 - a^2)^2 = -\frac{c}{2}(x^4 - 2a^2x^2 + a^4)$$

- **Symmetry:**

$$V(-x) = -\frac{c}{2}((-x)^2 - a^2)^2 = -\frac{c}{2}(x^2 - a^2)^2 = V(x) \implies \text{Even function (symmetric about } x = 0).$$

- **Critical Points & Equilibrium Configurations:**

Pedagogical Justification: Equilibrium points occur where the resultant force vanishes, $F(x) = -\frac{dV}{dx} = 0$.

$$\frac{dV}{dx} = -\frac{c}{2} \cdot 2(x^2 - a^2) \cdot (2x) = -2cx(x^2 - a^2)$$

Setting $\frac{dV}{dx} = 0$ yields three critical points:

$$x_1 = 0, \qquad x_2 = +a, \qquad x_3 = -a$$

- **Stability Evaluation ($V''(x)$):**

$$V''(x) = \frac{d}{dx}[-2cx(x^2 - a^2)] = \frac{d}{dx}[-2cx^3 + 2ca^2x] = -6cx^2 + 2ca^2 = 2c(a^2 - 3x^2)$$

1. At $x = 0$:

$$V(0) = -\frac{c}{2}(0 - a^2)^2 = -\frac{1}{2}ca^4$$

$$V''(0) = 2c(a^2 - 0) = +2ca^2 > 0 \implies \textbf{Local Minimum} \text{ (Stable Equilibrium)}$$

2. At $x = \pm a$:

$$V(\pm a) = -\frac{c}{2}(a^2 - a^2)^2 = 0$$

$$V''(\pm a) = 2c(a^2 - 3a^2) = -4ca^2 < 0 \implies \textbf{Local Maxima} \text{ (Unstable Equilibria / Potential Barriers)}$$

- **Asymptotic Limits:**

$$\lim_{x \to \pm\infty} V(x) = \lim_{x \to \pm\infty} \left[-\frac{c}{2}x^4\right] = -\infty$$

PHASE 3

Step-by-Step Mathematical Deduction

Pedagogical Justification: By examining the intersection of the horizontal energy line E with the graph of $V(x)$, we locate the turning points ($V(x) = E$) and delineate the allowed intervals where $T = E - V(x) \geq 0$.

Case (a): $E > 0$

- **Energy Relation:** The maximum value attained by the potential is $V_{max} = V(\pm a) = 0$. Since $E > 0$, we have:

$$E > V(x), \qquad \forall x \in (-\infty, +\infty)$$

- **Turning Points:** The equation $V(x) = E$ has no real solutions because $-\frac{c}{2}(x^2 - a^2)^2 \leq 0 < E$.
- **Kinetic Energy:**

$$T(x) = E - V(x) = E + \frac{c}{2}(x^2 - a^2)^2 > E > 0$$

The velocity never vanishes and never changes sign.

- **Type of Motion: Unbounded Motion.**
- A particle entering from $-\infty$ moves towards $+x$, accelerates as it enters the central region, passes through the barriers at $x = -a$ and $x = +a$ with speed $v \geq \sqrt{2E/m}$, and continues indefinitely to $+\infty$.
- Symmetrically, a particle entering from $+\infty$ with $v < 0$ traverses the entire axis to $-\infty$.

Case (b): $E < -\frac{1}{2}ca^4$

- **Energy Relation:** The energy is strictly below the local minimum at the origin, $E < V(0) = -\frac{1}{2}ca^4$.
- **Central Region:** For $|x| \leq a$, the minimum of the potential is $V(0) = -\frac{1}{2}ca^4 > E$. Therefore, in the central region:

$$T(x) = E - V(x) < 0$$

which is classically forbidden. The particle can **never** enter the region between the outer turning points.

- **Outer Turning Points:** Since $V(x) \to -\infty$ as $|x| \to \infty$, the energy line E intersects $V(x)$ at two points outside $[-a, a]$:

$$-\frac{c}{2}(x^2 - a^2)^2 = E \implies (x^2 - a^2)^2 = -\frac{2E}{c} = \frac{2|E|}{c}$$

Taking the positive square root:

$$x^2 - a^2 = \frac{2|E|}{c} \implies x^2 = a^2 + \frac{2|E|}{c}$$

$$x_t = \pm \sqrt{a^2 + \frac{2|E|}{c}} = \pm x_{out}, \qquad \text{with } x_{out} > a$$

- **Allowed Regions:** $x \in (-\infty, -x_{out}]$ and $x \in [+x_{out}, +\infty)$.
- **Type of Motion: Unbounded Motion with Reflection.**
- If the particle begins at $+\infty$ heading left ($v < 0$), it decelerates as $V(x)$ increases, stops momentarily at $x = +x_{out}$ ($T = 0$), and is accelerated back towards $+\infty$ by the positive restoring force $F = -V'(x_{out}) > 0$.
- Similarly, in the negative half-space, a particle from $-\infty$ reflects at $-x_{out}$ and returns to $-\infty$.

Case (c): $-\frac{1}{2}ca^4 < E < 0$

- **Energy Relation:** The energy level lies between the central local minimum and the two local maxima:

$$V(0) < E < V(\pm a)$$

- **Determination of Turning Points:**

$$-\frac{c}{2}(x^2-a^2)^2 = E \implies (x^2-a^2)^2 = \frac{2|E|}{c}$$

Because $|E| < \frac{1}{2}ca^4$, we have $\frac{2|E|}{c} < a^2$.

This produces two distinct positive solutions for x^2:

$$x^2 = a^2 \pm \frac{2|E|}{c}$$

yielding four real turning points:

$$x_{in} = a^2 - \frac{2|E|}{c} \in (0, a)$$

$$x_{out} = a^2 + \frac{2|E|}{c} > a$$

Ordered along the real axis:

$$-x_{out} < -x_{in} < 0 < x_{in} < x_{out}$$

- **Allowed Regions where $V(x) \leq E$:**

1. **Inner Well Region:** $x \in [-x_{in}, +x_{in}]$.

2. **Outer Left Region:** $x \in (-\infty, -x_{out}]$.

3. **Outer Right Region:** $x \in [+x_{out}, +\infty)$.

- **Type of Motion:**

- **Bounded Oscillatory Motion:** If the particle is placed inside the central well $[-x_{in}, +x_{in}]$, it is permanently confined by the potential barriers. It oscillates back and forth periodically between $-x_{in}$ and $+x_{in}$ around the stable equilibrium $x = 0$.
- **Unbounded Motion with Reflection:** If the particle begins in the outer regions ($|x| \geq x_{out}$), it approaches from $\pm\infty$, reaches the outer turning point $\pm x_{out}$, and is reflected back to $\pm\infty$.

PHASE 4

Physical Interpretation & Dimensional Verification

- **Summary Table:**

Energy Range	Turning Points	Allowed Regions	Motion Description
(a) $E > 0$	None	$(-\infty, +\infty)$	Unbounded: traverses from $-\infty$ to $+\infty$ (or vice versa).
(b) $E < -\frac{1}{2}ca^4$	$\pm x_{out}$	$(-\infty, -x_{out}] \cup [x_{out}, +\infty)$	Unbounded with reflection at $\pm x_{out}$ back to $\pm\infty$. Inner well forbidden.
(c) $-\frac{1}{2}ca^4 < E < 0$	$\pm x_{in}, \pm x_{out}$	$[-x_{in}, x_{in}]$ OR $\|x\| \geq x_{out}$	Bounded periodic oscillation in well $[-x_{in}, x_{in}]$, OR reflection at $\pm x_{out}$ back to $\pm\infty$.

- **Comparison with UC3M Solution Keys:** The qualitative classification can be compared with the official solution key (Problems.pdf, Solution keys section).
- **Dimensional check:** $[c\,a^4] = (\mathrm{J/m^4})\,\mathrm{m}^4 = \mathrm{J}$, the same as E; $[2|E|/c] = \mathrm{J}/(\mathrm{J/m^4}) = \mathrm{m}^4$, so $\sqrt{2|E|/c}$ is an area (m^2) and $x_{out} = \sqrt{a^2 + \sqrt{2|E|/c}}$ is a length.
- **Limiting energies (sympy):**
- $E \to -\frac{1}{2}ca^4$ (bottom of the well): $x_{in}^2 = a^2 - \sqrt{2|E|/c} \to 0$ and $x_{out}^2 \to 2a^2$, so the inner region collapses to the stable point $x = 0$, where the small oscillations have $\omega^2 = V''(0)/m = 2ca^2/m$.
- $E \to 0^-$ (top of the barriers): $x_{in}, x_{out} \to a$, the four turning points merge pairwise at the unstable equilibria $x = \pm a$ (separatrix between cases (c) and (a)).

Complete Step-by-Step Solution

Source: Problems.pdf (Page 9)